Notes A Functorial Excursion Between Algebraic Geometry and Linear Logic

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1 Preliminary Definitions

Definition 1.1. Let $\mathcal{C}$ be a category, then $\mathcal{C}$ is a Grothendieck category if the following is satisfied:

- $\mathcal{C}$ is Abelian, that is,
  - $\mathcal{C}$ has a zero object,
  - $\mathcal{C}$ has all binary biproducts,
  - $\mathcal{C}$ has all kernels and cokernels,

- all monomorphisms and epimorphisms are normal.

- $\mathcal{C}$ admits a generator, that is, there is $G \in \mathcal{C}$ such that for every parallel pair of morphisms $f, g : X \to Y$, $f \circ e = g \circ e$ implies $f = g$ for any $e : G \to X$.

- Let I be a directed set and let $0 \to A_i \to B_i \to C_i \to 0$ be an exact sequence for each $i \in I$, then $0 \to \operatorname{colim} A_i \to \operatorname{colim} B_i \to \operatorname{colim} C_i \to 0$ is an exact sequence.

Definition 1.2.

- Let $P : \mathcal{E} \to \mathcal{B}$ be a functor, then a morphism $f : A \to B$ in $\mathcal{E}$ is *Cartesian* if for any morphism $g : C \to B$ in $\mathcal{E}$ and $w : P(C) \to P(A)$ such that

$$\begin{tikzcd} P(C) \arrow[rr, "P(g)"] \arrow[dr, "w"'] & & P(B) \\ & P(A) \arrow[ur, "P(f)"'] & \end{tikzcd}$$

there exists a unique $\hat{w} : C \to A$ in $\mathcal{E}$ such that $g = f \circ \hat{w}$ and $P(\hat{w}) = w$:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "P"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} C \arrow[rr, "\forall g"] \arrow[dr, "\exists ! w"'] & & B \\ & A \arrow[ur, "f"'] & \end{tikzcd} \qquad \begin{tikzcd} P(C) \arrow[rr, "P(g)"] \arrow[dr, "\forall w"'] & & P(B) \\ & P(A) \arrow[ur, "P(f)"'] & \end{tikzcd}$$

- A functor $P : \mathcal{E} \to \mathcal{B}$ is a *Grothendieck fibration* if for all $A \in \mathcal{E}$ and morphisms $f_0 : B \to P(A)$ in $\mathcal{B}$ there is a Cartesian morphism $f : B \to A$ such that $P(f) = f_0$. $P : \mathcal{E} \to \mathcal{B}$ is an opfibration, if $P^{op} : \mathcal{E}^{op} \to \mathcal{B}^{op}$ is a fibration, and $P : \mathcal{E} \to \mathcal{B}$ is a bifibration if it is both a fibration and bifibration.

2 Motivational Thoughts

Let us start by recalling basic principles of algebraic geometry. The intuition is that a space X comes along with a topological space (X, Σ_X) such that every open set U is equipped with a set $\mathcal{O}_X(U)$ of local operations called *regular functions*. Such regular functions are typically defined as polynomials with division and each $\mathcal{O}_X(U)$ defines a commutative ring. In functorial terms, every space gives rise to a ringed space, that is, a topological space (X, Σ_X) equipped with a contravariant functor $\mathcal{O}_X : \Sigma_X^{op} \to \mathbf{Ring}$ from the set of opens Σ_X^{op} with the reversed inclusion order. The functorial action of $\mathcal{O}_X$ describes the restriction functions

$$\mathcal{O}_X(V \subseteq U) : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$$

which takes an operation $f \in \mathcal{O}_X(U)$ defined on an open set U and restrict it to an operation

$$f|_V := \mathcal{O}_X(V \subseteq U)(f)$$

on an open subset $V \subseteq U$. The functorial understanding of the concept of a space is supported by the observation that every commutative ring R defines a ringed space:

$$\operatorname{Spec} R = (\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$$

on the set of prime ideals $\mathfrak{p}$ of the commutative ring R. Such a ringed space is called the *affine scheme* associated to R. Every space X of the theory, called a *scheme*, is a patchwork of affine schemes $\operatorname{Spec} R_i$ glued together using the language of sheaf theory.

Every scheme X comes equipped with a symmetric monoidal category $\mathbf{qcMod}_X$ of quasicoherent sheaf of $\mathcal{O}_X$-modules. By a presheaf of $\mathcal{O}_X$-modules M of a ringed space $(X, \mathcal{O}_X)$, we mean a contravariant functor

$$M : \Omega_X^{op} \to \mathbf{Ab}$$

where $\mathbf{Ab}$ is the category of Abelian groups such that each Abelian group $M(U)$ defines a $\mathcal{O}_X(U)$-module. The Serre-Swann theorem states a vector bundle on a suitable ringed space $(X, \mathcal{O}_X)$ can be equivalently encoded as its sheaf of sections that defines a locally free and projective sheaf of $\mathcal{O}_X$-modules. A sheaf of $\mathcal{O}_X$ is called quasicoherent when it is locally presentable, that is, it is locally the cokernel of a morphism of free modules. The category $\mathbf{qcMod}_X$ extends the category of vector bundles to define an Abelian category where kernels and direct images can be computed.

The idea is to associate a model of intuitionistic linear logic, where formulas are understood as generalised vector bundles and proofs are vector fields by building an appropriate linear-non-linear adjunction. To associate a model of linear logic with each scheme X, we take the category $\mathbf{PshMod}_X$ of presheafs of $\mathcal{O}_X$-modules and their homomorphisms as the underlying symmetric monoidal closed category with tensor product $\otimes_X$, the structural sheaf $\mathcal{O}_X$ as unit, and $\multimap_X$ for internal homs. For the cartesian part of a linear-non-linear adjunction, we take the category $\mathbf{PshCoAlg}_X$ of cocommutative comonoids (or commutative coalgebras) in $\mathbf{PshMod}_X$. It is convenient to the approach based on the functor of points (see [EH00, I.4]). Spec defines a contravariant functor $\operatorname{Spec} : \mathbf{Ring}^{op} \to \mathbf{Scheme}$ that induces the *nerve functor*

$$\mathbf{nerve} : \mathbf{Scheme} \to [\mathbf{Ring}, \mathbf{Set}]$$

associating to every commutative ring R *the set of R-points in a scheme X* defined as $\mathbf{Scheme}(\operatorname{Spec} R, X)$. The nerve functor is fully faithful and it allows one to see every scheme X as a covariant presheaf

$$\mathbf{nerve}(X) : \mathbf{Ring} \to \mathbf{Set}$$

So we move to schemes to $\mathbf{Ring}$-spaces and associate a model of intuitionistic linear logic with each $\mathbf{Ring}$-space X, where formulas are interpreted in

$\mathbf{PshMod}_X$ as presheaves of $\mathcal{O}_X$-module understood as generalised vector bundles on the **Ring**-space. The linear-non-linear adjunction (and therefore the exponential modality) is constructed by the Sweedler construction.

3 On the category of rings

Let $(\mathcal{A}, \otimes, \mathbb{1})$ be a symmetric monoidal category with reflexive coequalisers preserved by tensor products in every coordinate. For example, the category **Ab** of Abelian groups and homomorphisms. A ring is a monoid object (R, m, e) in $(\mathcal{A}, \otimes, \mathbb{1})$. In other words, a ring is a triple such that (R, m, e) consisting of an object R and morphisms $m : R \otimes R \to R$ and $e : \mathbb{1} \to R$ such that the following diagrams commute:

$$
\begin{tikzcd}
R \otimes R \otimes R \arrow[r, "m\otimes R"] \arrow[d, "R\otimes m"'] & R \otimes R \arrow[d, "m"] \\
R \otimes R \arrow[r, "m"'] & R
\end{tikzcd}
\qquad
\begin{tikzcd}
R \otimes R \arrow[r, "m"] \arrow[d, "m"'] & R \arrow[d, "e\otimes R"] \\
R \arrow[ur, equal] \arrow[r, "R\otimes e"'] & R \otimes R
\end{tikzcd}
$$

A ring (R, m, e) is commutative is the following triangle commutes:

$$
\begin{tikzcd}
R \otimes R \arrow[rr, "\gamma_{R,R}"] \arrow[dr, "m"'] & & R \otimes R \arrow[dl, "m"] \\
& R &
\end{tikzcd}
$$

A *ring morphism* $u : (R, m_R, e_R) \to (S, m_S, e_S)$ is a morphism $u : R \to S$ in the underlying category $\mathcal{A}$ making the following diagrams commute:

$$
\begin{tikzcd}
R \otimes R \arrow[r, "u\otimes u"] \arrow[d, "m_S"'] & S \otimes S \arrow[d, "m_S"] \\
R \arrow[r, "u"'] & S
\end{tikzcd}
\qquad
\begin{tikzcd}
& \mathbb{1} \arrow[dl, "e_R"'] \arrow[dr, "e_S"] & \\
R \arrow[rr, "u"'] & & S
\end{tikzcd}
$$

The category $\mathbf{Ring}(\mathcal{A})$ of rings in $\mathcal{A}$ defined by the tensor product $R, S \mapsto R \otimes S$ in $\mathcal{A}$.

4 On the category of R-modules and modules

Given a ring R in a symmetric monoidal category $\mathcal{A}$, an *R-module* is a pair (M, act) where M is an object in $\mathcal{A}$ along with a morphism $\mathrm{act} : R \otimes M \to M$ in $\mathcal{A}$ such that:

$$
\begin{tikzcd}
R \otimes R \otimes M \arrow[r, "m_R\otimes M"] \arrow[d, "R\otimes \mathrm{act}"'] & R \otimes M \arrow[d, "\mathrm{act}"] \\
R \otimes M \arrow[r, "\mathrm{act}"'] & M
\end{tikzcd}
\qquad
\begin{tikzcd}
& R \otimes M \arrow[dr, "\mathrm{act}"] & \\
M \arrow[ur, "e_R\otimes M"] \arrow[rr, equal] & & M
\end{tikzcd}
$$

Equivalently, an R-module is an Eilenberg-Moore algebra for the monad $A \mapsto R \otimes A$. An R-module homomorphism $f : M \to N$ between two modules is a map f in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{tikzcd} R \otimes M \arrow[r, "R\otimes f"] \arrow[d, "\mathrm{act}_M"'] & R \otimes N \arrow[d, "\mathrm{act}_N"] \\ M \arrow[r, "f"'] & N \end{tikzcd}$$

$\mathbf{Mod}_R$ is the category of R-modules and their morphisms.

A *module* is a pair (R, M) where R is a ring and M is a module. A *module homomorphism* $(u, f) : (R, M) \to (S, N)$ is a ring morphism $u : R \to S$ and a morphism $f : M \to N$ such that

$$\begin{tikzcd} R \otimes M \arrow[r, "u\otimes f"] \arrow[d, "\mathrm{act}_M"'] & S \otimes N \arrow[d, "\mathrm{act}_N"] \\ M \arrow[r, "f"'] & N \end{tikzcd}$$

$\mathbf{Mod}$ is the category of modules and their morphisms.

There is a functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ transporting every module (R, M) to the underlying ring R and every morphism (u, f) to u. We will write

$$u : R \to S \models f : M \to N$$

for a module morphism $(u, f) : (R, M) \to (S, N)$. The intuition behind the notation is that every ring homomorphism $u : R \to S$ induces a "fibre" consisting of all module morphisms $(u, f) : (R, M) \to (S, N)$ living above and refining the ring morphism $u : R \to S$. The fibre of πR for a commutative ring is the category $\mathbf{Mod}_R$. The functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ defines a Grothendieck fibration: every pair consisting of a ring morphism $u : R \to S$ and of an S-module (N, act_N) induces an R-module noted

$$\mathbf{res}_u N = (N, \mathrm{act}'_N)$$

with the action morphism act'_N defined as the composite:

$$R \otimes N \xrightarrow{u\otimes N} S \otimes N \xrightarrow{\mathrm{act}_N} N$$

Moreover, the S-module with a module morphism:

$$u : R \to S \models id_N : \mathbf{res}_u N \to N$$

which is weakly Cartesian in the sense that every module morphism

$$u : R \to S \models f : M \to N$$

factors uniquely as

$$M \xrightarrow{(id_R, h)} \mathbf{res}_u N \xrightarrow{(u, id_N)} N$$

for a morphism $h : M \to \mathbf{res}_u N$ in the category $\mathbf{Mod}_R$. The unique solution h is equal to the original morphism $f : M \to N$ viewed as a morphism in $\mathbf{Mod}_R$. Given a ring morphism $u : R \to S$, the existence of a weakly Cartesian map for every S-module N ensures the existence of a functor

$$\mathbf{res}_u : \mathbf{Mod}_S \to \mathbf{Mod}_R$$

called restriction of scalar along u. Along with the fact that the composite of weakly Cartesian maps is weakly Cartesian we have that the forgetful functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ is a Grothendieck fibration.

Now we use that fact the category $\mathcal{A}$ has reflexive coequalisers, so one can show that $\mathbf{res}_u$ has a left adjoint

$$\mathbf{ext}_u : \mathbf{Mod}_R \to \mathbf{Mod}_S$$

constructed as follows. Every ring homomorphism $u : R \to S$ induces an $R \otimes S$-module notes as $R \otimes_u S$ defined as the reflexive coequaliser of the following diagram

$$\begin{tikzcd} R \otimes R \otimes S \arrow[r, shift left=2, "m_R \otimes S"] \arrow[r, shift right=2, "(R \otimes m_S) \circ (R \otimes u \otimes S)"'] & R \otimes S \arrow[l, "R \otimes e_R \otimes S" description] \end{tikzcd}$$

computed in $\mathcal{A}$. Given three commutative rings R, S_1 and S_2, we define the functor

$$\circledast_R : \mathbf{Mod}_{S_1 \otimes R} \times \mathbf{Mod}_{R \otimes S_2} \to \mathbf{Mod}_{S_1 \otimes S_2}$$

The two maps $\mathrm{act}_M : M \otimes R \to M$ and $\mathrm{act}_N : N \otimes R \to N$ are deduced by restriction of scalar from the $S_1 \otimes R$-module structure of M and the $R \otimes S_2$-module structure of N. The left adjunction functor $\mathbf{ext}_u$ is then defined the following way:

$$\mathbf{ext}_u : M \mapsto M \circledast_R (R \otimes_u S)$$

5 The category $\mathbf{Mod}^{\ominus}$ of modules and retromorphisms

Let $[M, N]$ denote the internal hom of M and N in $\mathcal{A}$. A *module retromorphism*

$$(u, f) : (S, N) \to (R, M)$$

is a pair (u, f) where $u : R \to S$ is a ring morphism and $f : N \to M$ is a morphism in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{tikzcd} R \otimes M \arrow[d, "\mathrm{act}_M"'] & R \otimes N \arrow[l, "R \otimes f"'] \arrow[r, "u \otimes N"] & S \otimes N \arrow[d, "\mathrm{act}_N"] \\ M & & N \arrow[ll, "f"] \end{tikzcd}$$

The category $\mathbf{Mod}^{\ominus}$ has ring modules as objects and module retromorphisms. There is a forgetful functor $\pi : \mathbf{Mod}^{\ominus} \to \mathbf{Ring}$ transporting every retromorphism to the underlying ring morphism and π is also a Grothendieck fibration. π is also a bifibration. Let $u : R \to S$ be a ring morphism, then functor defined by coextension of scalar along u

$$\mathbf{coext}_u : \mathbf{Mod}_R \to \mathbf{Mod}_S$$

transport every R-module (M, act_M) to the S-module $[S, M]_u$ defined as the coreflexive equaliser of the below diagram:

$$\begin{tikzcd} {[S,M]} \arrow[r, shift left=2, "{[u\otimes S,M]\circ[m_S,M]}"] \arrow[r, shift right=2, "{[R\otimes S,\mathrm{act}_M]\circ[R\otimes\_,R\otimes\_]}"'] & {[R\otimes S, M]} \arrow[l, "{[e_R\otimes M,M]}" description] \end{tikzcd}$$

Note that $[S, M]_u$ provides an internal description of the set of morphisms $f : S \to M$ making the following diagram commute:

$$\begin{tikzcd} R\otimes S \arrow[r, "R\otimes f"] \arrow[d, "u\otimes S"'] & R\otimes M \arrow[dd, "\mathrm{act}_M"] \\ S\otimes S \arrow[d, "m_S"'] & \\ S \arrow[r, "f"'] & M \end{tikzcd}$$

or, equivalently, as the set of morphisms $f : \mathbf{res}_u S \to M$. Thus, every ring morphism $u : R \to S$ induces the following sequence of adjunctions:

$$\begin{tikzcd} \mathbf{Mod}_R \arrow[r, shift left=3, "\mathbf{coext}_u"] \arrow[r, shift right=3, "\mathbf{ext}_u"'] & \mathbf{Mod}_S \arrow[l, "\mathbf{res}_u" description] \end{tikzcd}$$

6 The category $\mathbf{Mod}_R$ is symmetric monoidal closed

It is well known that the category $\mathbf{Mod}_R$ of modules over a ring R is symmetric monoidal closed. Its tensor product

$$\otimes_R : \mathbf{Mod}_R \times \mathbf{Mod}_R \to \mathbf{Mod}_R$$

transports every pair of R-modules (M, N) into the coreflexive equaliser of the following diagram calculated in $\mathcal{A}$:

$$\begin{tikzcd} M\otimes R\otimes N \arrow[r, shift left=2, "\mathrm{act}_M\otimes N"] \arrow[r, shift right=2, "M\otimes\mathrm{act}_N"'] & M\otimes N \arrow[l, "M\otimes e_R\otimes N" description] \end{tikzcd}$$

In $\mathbf{Mod}_R$, the tensorial unit is the R itself seen as an R-module. The internal hom

$$[.,.]_R : \mathbf{Mod}_R^{op} \times \mathbf{Mod}_R \to \mathbf{Mod}_R$$

takes every pair of R-modules M and N to the coreflexive equaliser of the following diagram:

$$\begin{tikzcd}
{[M,N]} \arrow[r, shift left=2, "{[\mathrm{act}_M, N]}"] \arrow[r, shift right=2, "{[R\otimes M, \mathrm{act}_N]\circ[R\otimes_, R\otimes_]}"'] & {[R\otimes M, N]} \arrow[l, "{[e_R\otimes M, N]}" description]
\end{tikzcd}$$

7 The category Mod as a symmetric monoidal ringed category

Let us consider the category $\mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod}$ defined by the pullback diagram (in $\mathbf{Cat}$?) as follows:

$$\begin{tikzcd}
\mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} \arrow[r] \arrow[d] & \mathbf{Mod} \arrow[d, "\pi"] \\
\mathbf{Mod} \arrow[r, "\pi"'] & \mathbf{Ring}
\end{tikzcd}$$

The fibrewise tensor product

$$\otimes_{\mathbf{Mod}} : \mathbf{Mod} \times_{\mathbf{Ring}} \mathbf{Mod} \to \mathbf{Mod}$$

transports every pair of modules (R, M) and (R, N) on the same monoid R to the R-module $(R, M \otimes_R N)$, so every pair of module homomorphisms:

$$u : R \to S \models h_1 : M_1 \to N_1$$
$$u : R \to S \models h_2 : M_2 \to N_2$$

above the same ring morphism $u : R \to S$ to the module homomorphism:

$$u : R \to S \models h_1 \otimes_u h_2 : M_1 \otimes_R M_2 \to N_1 \otimes_R N_2$$

where $h_1 \otimes_u h_2$ is a unique morphism making the following diagram commute:

$$\begin{tikzcd}
M_1 \otimes R \otimes M_2 \arrow[r, "h_1\otimes u\otimes h_2"] \arrow[d, shift right=2, "M_1\otimes \mathrm{act}_{M_2}"'] \arrow[d, shift left=2, "\mathrm{act}_{M_1}\otimes M_2"] & N_1 \otimes S \otimes N_2 \arrow[d, shift right=2, "N_1\ \mathrm{act}_{N_2}"'] \arrow[d, shift left=2, "\mathrm{act}_{N_1}\otimes N_2"] \\
M_1 \otimes M_2 \arrow[r, "h_1\otimes h_2"] \arrow[d, "quotient"'] & N_1 \otimes N_2 \arrow[d, "quotient"] \\
M_1 \otimes_R M_2 \arrow[r, "h_1\otimes_u h_2"'] & N_1 \otimes_S N_2
\end{tikzcd}$$

The fibrewise unit $\mathbb{1}_{\mathbf{Mod}} : \mathbf{Ring} \to \mathbf{Mod}$ takes every commutative ring and converts it to itself viewed as a module. Categorically, one can understood that with the concept of a *ringed category*, which defined as a pair $(\mathcal{C}, \pi)$ where $\mathcal{C}$ is a category and $\pi : \mathcal{C} \to \mathbf{Ring}$ is a functor. The slice 2-category $\mathcal{C}/\mathbf{Ring}$ consists of ringed categories, fibrewise functions as 1-cells and natural transformations as 2-cells. The 2-category $\mathcal{C}/\mathbf{Ring}$ is Cartesian; the fibrewise product and the unit define a symmetric pseudomonoid structure on the ringed category $(\mathbf{Mod}, \pi)$ in $\mathcal{C}/\mathbf{Ring}$.

8 The categories Alg and CoAlg of (co)commutative (co)algebras

Let R be a commutative ring in $\mathcal{A}$, a *commutative R-algebra* is a commutative monoid in the symmetric monoidal category $\mathbf{Mod}_R$. A *commutative algebra* is a pair (R, A) where R is a commutative ring and A is a commutative R-algebra. An algebra morphism $(u, f) : (R, A) \to (S, B)$ is a pair (u, f) where u is a ring morphism and $f : A \to B$ is a module morphism such that the following diagrams commute:

$$\begin{tikzcd} A \otimes_R A \arrow[r, "f \otimes_u f"] \arrow[d, "m_A"'] & B \otimes_S B \arrow[d, "m_B"] \\ A \arrow[r, "f"'] & B \end{tikzcd} \qquad \begin{tikzcd} 1_R \arrow[r, "u"] \arrow[d, "e_A"'] & 1_S \arrow[d, "e_B"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

Assume that for every commutative ring R the symmetric monoidal category $\mathbf{Mod}_R$ has free commutative monoids, which means that the forgetful functor

$$\mathbf{Forget}_R : \mathbf{Alg}_R \to \mathbf{Mod}_R$$

has a left adjoint

$$\mathbf{Sym}_R : \mathbf{Mod}_R \to \mathbf{Alg}_R$$

and the adjunction $\mathbf{Mod} \dashv \mathbf{Alg}$ is also fibrewise above $\mathbf{Ring}$.

Given a commutative ring R, a *cocommutative R-coalgebra K* is defined as a cocommutative comonoid in the symmetric monoidal category $\mathbf{Mod}_R$. Let us also assume that for any commutative ring R, $\mathbf{Mod}_R$ has cofree cocommutative comonoids, then the forgetful functor

$$\mathbf{Forget}_R : \mathbf{CoAlg}_R \to \mathbf{Mod}_R$$

has a right adjoint

$$\mathbf{Sym}_R : \mathbf{Mod}_R \to \mathbf{CoAlg}_R.$$

The above adjunction can be defined at a purely formal level without describing explicitly the cofree construction $\mathbf{CoFree}_R$ in each fibre $\mathbf{Mod}_R$. One can describe the construction of the above adjunction explicitly in many cases, for example when $R = k$ is an algebraic closed field, see [Mur15]. One can also describe it for a general commutative ring R. First of all, the free commutative algebra $\mathbf{Sym}_R$ is calculated the following way with enough limits in the category $\mathbf{Mod}_R$:

$$\mathbf{Sym}_R : M \mapsto \bigoplus_{n<\omega} M \otimes_R^{sym} \ldots \otimes_R^{sym} M$$

where $M \otimes_R^{sym} \ldots \otimes_R^{sym} M$ is the symmetrised tensor product of M with itself n times. If $R = k$ is a field, then the Sweedler construction transports every k-vector space V into the commutative k-algebra defined the below way:

$$\mathbf{CoFree}_k(V) = (\mathbf{Sym}_k(V^*))^\circ$$

where V^* is the dual of V and $A \mapsto A^\circ$ is the Sweedler finite dual constuction mapping every R-algebra A into the R-coalgebra A°. In the case of a general commutative ring, one needs to use the construction from [Por15].

9 Functor of Points, Ring-spaces and Presheaves of Modules

We work with covariant presheaves X, Y on the category **Ring** of commutative rings that we call **Ring**-*spaces*. To every such space $X : \mathbf{Ring} \to \mathbf{Set}$ we associate its Grothendieck category $\mathbf{Points}(X)$ whose objects are pairs (R, x) with $x \in X(R)$ and whose morphisms $(R, x) \to (S, y)$ are ring morphisms $u : R \to S$ transporting the element $x \in X(R)$ to the element $y \in X(S)$ in the sense that $X(u)(x) = y$.

The category $\mathbf{Points}(X)$ comes with an obvious functor $\pi_X : \mathbf{Points}(X) \to \mathbf{Ring}$ such that $\pi_X : (R, x) \mapsto R$. π_X is called the functor of points of X and it defines a discrete Grothendieck opfibration above the category **Ring**. A morphism $f : X \to Y$ between **Ring**-spaces may be equivalently defined as a functor:

$$f : \mathbf{Points}(X) \to \mathbf{Points}(Y)$$

making the following diagram commute:

$$\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "f"] \arrow[dr, "\pi_X"'] & & \mathbf{Points}(Y) \arrow[dl, "\pi_Y"] \\
& \mathbf{Ring} &
\end{tikzcd}$$

The functor f is necessarily a discrete opfibration. The following **Ring**-space

$$\operatorname{Spec} \mathbb{Z} : R \mapsto \{*_R\}$$

is the terminal object in the presheaf category $[\mathbf{Ring}, \mathbf{Set}]$, and that its Grothendieck category is isomorphism to **Ring**.

The following definition is adapted from [Ros11].

Definition 9.1. A *presheaf of modules M on the* **Ring**-*space* or an $\mathcal{O}_X$*-module* M consists of the following data:

- For each point $(R, x) \in \mathbf{Points}(X)$, a module M_x over the commutative ring R,
- Each map $u : (R, x) \to (S, y)$ in $\mathbf{Points}(X)$, a module morphism

$$u : R \to S \models \theta(u, x) : M_x \to N_y$$

lining over the ring morphism $u : R \to S$.

The map θ is required to satisfy the following functorial properties: the identity on the point (R, x) in $\mathbf{Points}(X)$ is transported to the identity morphism on the associated R-module

$$1_R \models \theta(1_{(R,x)}) = 1_{M_x}$$

and given two maps $(u, x) : (R, x) \to (S, y)$ and $(v, y) : (S, y) \to (T, z)$ in $\mathbf{Points}(X)$, one has:

$$v \circ u \models \theta((v, y) \circ (u, x)) = \theta(v, y) \circ \theta(u, x)$$

where the composite is calculated in the discrete opfibration $\mathbf{Points}(X) \to \mathbf{Ring}$.

There is the following characterisation:

Proposition 9.1. An $\mathcal{O}_X$-module M is the same thing as a functor $M : \mathbf{Points}(X) \to \mathbf{Mod}$ making the following diagram commute:

$$\begin{tikzcd} \mathbf{Points}(X) \arrow[rr, "M"] \arrow[dr, "\pi_X"'] & & \mathbf{Mod} \arrow[dl, "\pi"] \\ & \mathbf{Ring} & \end{tikzcd}$$

Note that every $\mathbf{Ring}$-every space X comes with a presheaf of modules called *the structure presheaf* of X, and defined as the composite

$$\mathcal{O}_X : \mathbf{Points}(X) \xrightarrow{\pi_X} \mathbf{Ring} \xrightarrow{i_{\mathbf{Ring}}} \mathbf{Mod}$$

where the functor $i_{\mathbf{Ring}} : \mathbf{Ring} \to \mathbf{Mod}$ is the section of the forgetful functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ transporting every ring to itself viewed as a module.

10 The category PshMod of Presheaves of modules and forward morphisms

We construct the category **PshMod** of *presheaves of modules and forward morphisms as follows.* A *presheaf of modules* (X, M) is defined as a pair consisting of a **Ring**-space $X : \mathbf{Ring} \to \mathbf{Set}$ along with a presheaf $\mathcal{O}_X$-module M as above.

A *forward morphism* between presheafs of modules $(f, \varphi) : (X, M) \to (Y, N)$ is defined as a morphism $f : X \to Y$ along with a natural transformation:

$$\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[dr, "M"'] & {} \arrow[d, phantom, "\overset{\varphi}{\Longrightarrow}"] & \mathbf{Points}(Y) \arrow[dl, "N"] \\
& \mathbf{Mod} &
\end{tikzcd}$$

φ is also required to vertical above **Ring**, i.e. the natural transformation obtained by composing

$$\pi \circ M \xRightarrow{\pi \circ \varphi} \pi \circ N \circ \mathbf{Points}(f)$$

is equal to the identity natural transformation. Equivalently, one can say that the following natural transformation

$$\pi_X \xRightarrow{1} \pi \circ M \xRightarrow{\pi \circ \varphi} \pi \circ N \circ \mathbf{Points}(f) \xRightarrow{1} \pi_Y \circ \mathbf{Points}(f)$$

is obtained by composing the three natural transformations depicted below:

$$\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[dr, "M"'] \arrow[ddr, bend right, "\pi_X"'] & {} \arrow[d, phantom, "\overset{\varphi}{\Longrightarrow}"] & \mathbf{Points}(Y) \arrow[dl, "N"] \arrow[ddl, bend left, "\pi_Y"] \\
{} \arrow[r, phantom, "\Uparrow \mathrm{id}"] & \mathbf{Mod} \arrow[d, "\pi"] \arrow[r, phantom, "\Downarrow \mathrm{id}"] & {} \\
& \mathbf{Ring} &
\end{tikzcd}$$

coincides with the identity natural transformation from π_X to $\pi_Y \circ f$. There is an obvious functor

$$\mathbf{p} : \mathbf{PshMod} \to [\mathbf{Ring}, \mathbf{Set}]$$

that transports every presheaf of modules (X, M) to its underlying **Ring**-space X and every forward morphism $(f, \varphi) : (X, M) \to (Y, N)$ to $f : X \to Y$ between **Ring**-spaces. We will further write

$$f : X \to Y \models \varphi : M \to N$$

for a forward morphirm $(f, \varphi) : (X, M) \to (Y, N)$ between presheaf modules.

The functor $\mathbf{p}$ is a Grothendieck fibration every **Ring**-space morphism $f : X \to Y$ induces a functor

$$f^* : \mathbf{PshMod}_Y \to \mathbf{PshMod}_X$$

transporting every $\mathcal{O}_Y$-module N into the $\mathcal{O}_X$-module $N \circ \mathbf{Points}(f)$ obtained by precomposing with the discrete fibration $\mathbf{Points}(f)$ as depicted below:

$$\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[ddr, "\pi_X"'] & & \mathbf{Points}(Y) \arrow[dl, "N"'] \arrow[ddl, "\pi_Y"] \\
& \mathbf{Mod} \arrow[d, "\pi"] & \\
& \mathbf{Ring} &
\end{tikzcd}$$

$\mathbf{p}$ is also a Grothendieck bifibration, but it is less straightforward. Let us also assume that $\mathbf{Ring}$ as well as $\mathbf{Mod}_R$ has small colimits. Such a property is the case for, say, $\mathcal{A} = \mathbf{Ab}$.

Proposition 10.1. For every morphism $f : X \to Y$ of $\mathbf{Ring}$-spaces, there is a functor

$$f_! : \mathbf{PshMod}_X \to \mathbf{PshMod}_Y$$

which is left adjoint to f^*.

Proof. A proof of the statement based on a 2-categorical construction of the $\mathcal{O}_Y$-module $f_!(M)$. One uses the fact that the functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ is a Grothendieck bifibration, in particular, a Grothendieck opfibration. This implies that $\mathbf{Mod}$ has small colimits, which are preserved by π. Suppose we are given the following diagram:

$$\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[dr, "M"] \arrow[ddr, "\pi_X"'] & & \mathbf{Points}(Y) \arrow[ddl, "\pi_Y"] \\
& \mathbf{Mod} \arrow[d, "\pi"] & \\
& \mathbf{Ring} &
\end{tikzcd}$$

We start by calculating the left Kan extension of the functor M along the discrete fibration

$$\mathbf{Points}(f) : \mathbf{Points}(X) \to \mathbf{Points}(Y)$$

One further obtains a functor $M' : \mathbf{Points}(Y) \to \mathbf{Mod}$ and a natural transformation λ exhibiting M' as the left Kan extension of M along points $\mathbf{Points}(f)$ depicted below:

$$\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[dr, "M"', ""{name=A}] \arrow[ddr, "\pi_X"'] & & \mathbf{Points}(Y) \arrow[dl, "M'", ""'{name=B}] \arrow[ddl, "\pi_Y"] \\
& \mathbf{Mod} \arrow[d, "\pi"] & \\
& \mathbf{Ring} &
\arrow[Rightarrow, from=A, to=B, "\lambda"]
\end{tikzcd}$$

The left Kan extension is a pointwise Kan extension calculated by small limits. As mentioned above, π preserves small colimits, and thus pointwise left Kan extensions. This established that the natural transformation $\pi \circ \lambda$ exhibits the functor $\pi \circ N'$ as the left Kan extension of the composite $\pi \circ M$ along $\mathbf{Points}(f)$. The universality property of left Kan extensions ensures that there exists a unique natural transformation $\mu : \pi \circ N' \Rightarrow \pi_Y$:

$$
\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[dr, "M"] \arrow[ddr, "\pi_X"'] \arrow[rr, phantom, shift right=6ex, "\overset{\lambda}{\Longrightarrow}"] & & \mathbf{Points}(Y) \arrow[dl, "M'"'] \arrow[ddl, "\pi_Y"] \arrow[ddl, phantom, shift right=2ex, "\Downarrow\mu"] \\
& \mathbf{Mod} \arrow[d, "\pi"] & \\
& \mathbf{Ring} &
\end{tikzcd}
$$

such that the following composite:

$$\pi \circ M \xRightarrow{\pi\circ\lambda} \pi \circ M' \circ \mathbf{Points}(f) \xRightarrow{\mu\circ\mathbf{Points}(f)} \pi_Y \circ \mathbf{Points}(f)$$

is equal to the identity natural transformation on the functor $\pi_X = \pi_Y \circ \mathbf{Points}(f)$. The functor π is a Grothendieck bifibration, so is the associated postcomposition functor:

$$\mathbf{Cat}(\mathbf{Points}(Y), \mathbf{Mod}) \to \mathbf{Cat}(\mathbf{Points}(Y), \mathbf{Ring})$$

is also a Grothendieck bifibration. In particular, there exists a functor $N : \mathbf{Points}(Y) \to \mathbf{Mod}$ and a natural transformation $\nu : M \Rightarrow N'$, which is cocartesian above the natural transformation $\mu : \pi \circ M' \Rightarrow \pi \circ N$ as in the diagram below:

$$
\begin{tikzcd}
\mathbf{Points}(X) \arrow[rr, "\mathbf{Points}(f)"] \arrow[dr, "M"] \arrow[ddr, bend right, "\pi_X"'] \arrow[rr, phantom, shift right=6ex, "\overset{\lambda}{\Longrightarrow}"] & & \mathbf{Points}(Y) \arrow[dl, bend right, "M'"'] \arrow[dl, bend left, "N'"] \arrow[dl, phantom, "\overset{\nu}{\Rightarrow}"] \arrow[ddl, bend left, "\pi_Y"] \\
& \mathbf{Mod} \arrow[d, "\pi"] & \\
& \mathbf{Ring} &
\end{tikzcd}
$$

The natural transformation $\varphi = \nu \circ \lambda : M \Rightarrow N \circ \mathbf{Points}(f)$ is vertical. Besides, $\pi \circ \varphi = 1$ and $N \circ \pi = \pi_Y$. The $\mathcal{O}_X$-module N is defined $N : \mathbf{Points}(Y) \to \mathbf{Ring}$. Therefore we obtain a functor $f_! : \mathbf{PshMod}_X \to \mathbf{PshMod}_Y$ is left adjoint to the inverse image functor $f^* : \mathbf{PshMod}_Y \to \mathbf{PshMod}_X$ defined earlier.

$\square$

Note that the $\mathcal{O}_Y$-module $f_!(M)$ can be explicitly described by the following formula:

$$f_!(M) : y \in Y(R) \mapsto \bigoplus_{x \in X(R), fx = y} M_x \in \mathbf{Mod}_R$$

The adjunction $f_! \dashv f^*$ gives rise to a sequence of natural bijections formulated as the following derivations:

$$\cfrac{\cfrac{1_X : X \to X \models M \to f^*(N)}{f : X \to Y \models M \to N}}{1_Y : Y \to Y \models f_!(M) \to N}$$

11 The category $\mathbf{PshMod}^{\ominus}$ of presheaf modules and backward morphisms

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